Case Study **4**:

Analyzing effects of the Federal Reserve's Monetary Tightening

NIKITHA JOSHY

Problem Introduction

- Federal Reserve Bank of St. Louis
- FRED Database
- Monetary Tightening (2021–2025)
  - Pre-Trend
  - Shock Window
  - Transmission Period
- How did this Affect:
  - Auto Credit & Loan Outcomes
  - Consumer Credit & Bank Lending
  - Manufacturing Output

Objective: Examine how Fed's 2021–2025 monetary tightening affected credit and manufacturing sectors.

Features from FRED Used in Analysis



General



Auto



Credit



Manufacturing

Assumptions + Observations

- Pre-Trend: Jan 2018 – Mar 2020
- Shock Window: Mar 2020 – Apr 2020
- Transmission: May 2020 – Dec 2020

- Unique Observations
  - Most data was recorded monthly.
  - Quarterly series were converted to monthly for consistent analysis.
  - Had to normalize & standardize data.

Approach



- Correlation & Time-Series Analysis.
- Focused on the 3 main Effects.
- Used visual trend analysis and correlation values to interpret results.

Findings

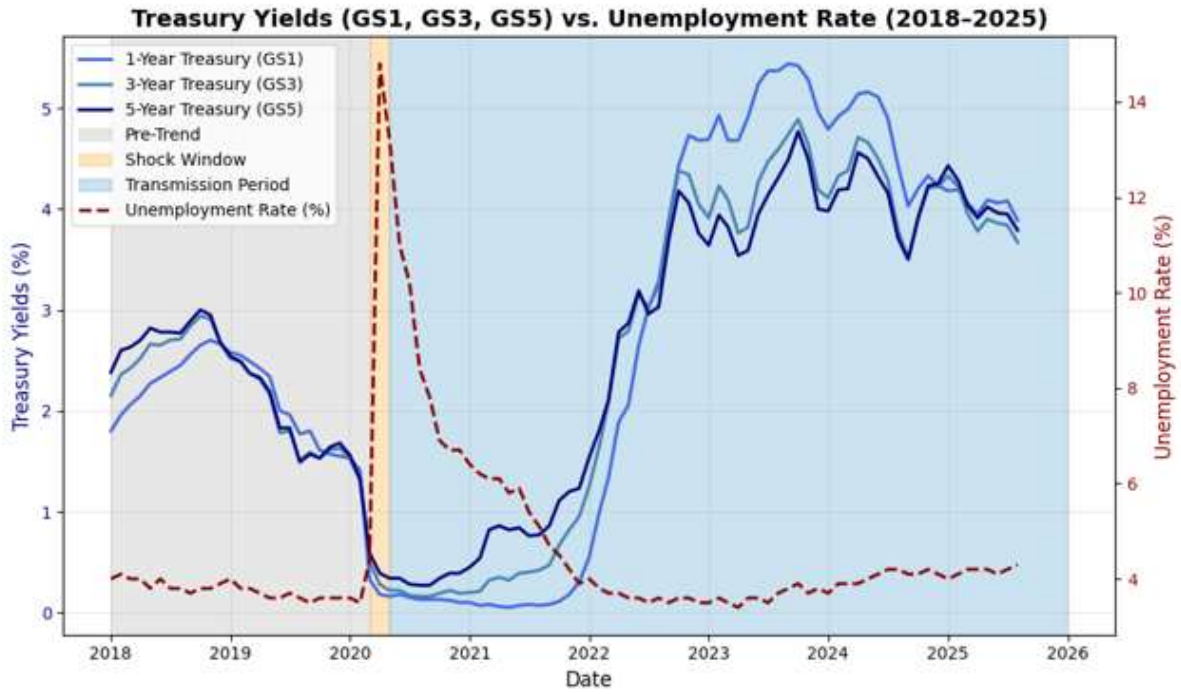


Figure 7. Treasury Yields and Unemployment Rate

- Unemployment Rate against Treasury Yields.

Auto Credit & Loan Outcomes

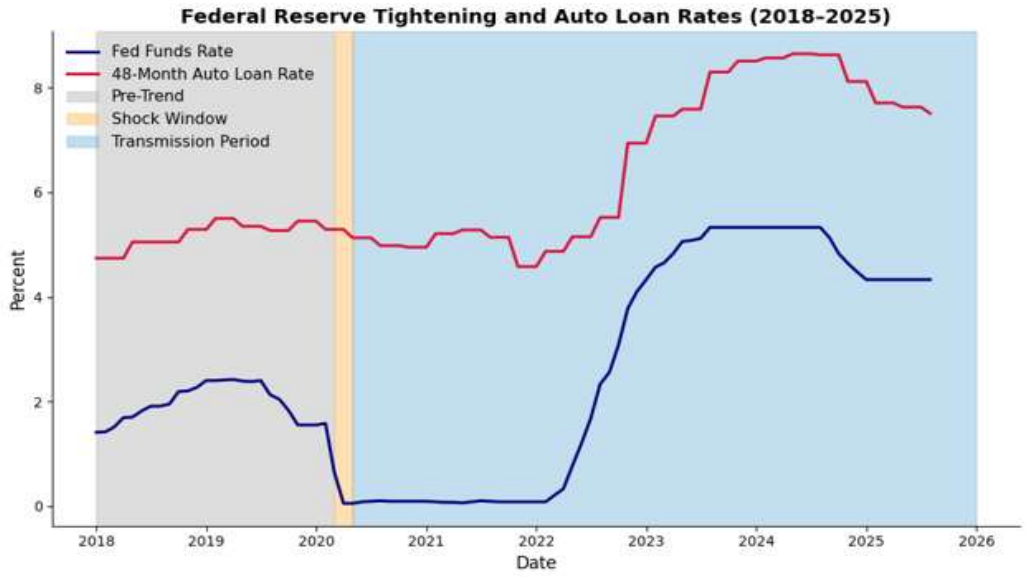


Figure 1. Federal Reserve Tightening and Auto Loan Rates (2018–2025)

- How the Federal Reserve tightening changed along with the 48-Month Auto Loan Rate

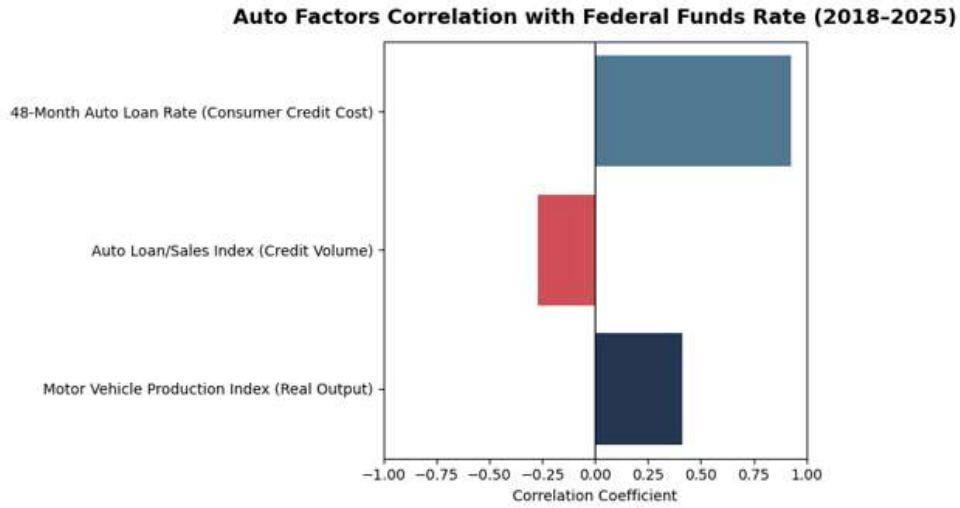


Figure 4. Auto Factors Correlation with Fed Funds

- Visual representation of how different Auto Factors correlate with Federal Funds.

Consumer Credit & Bank Lending

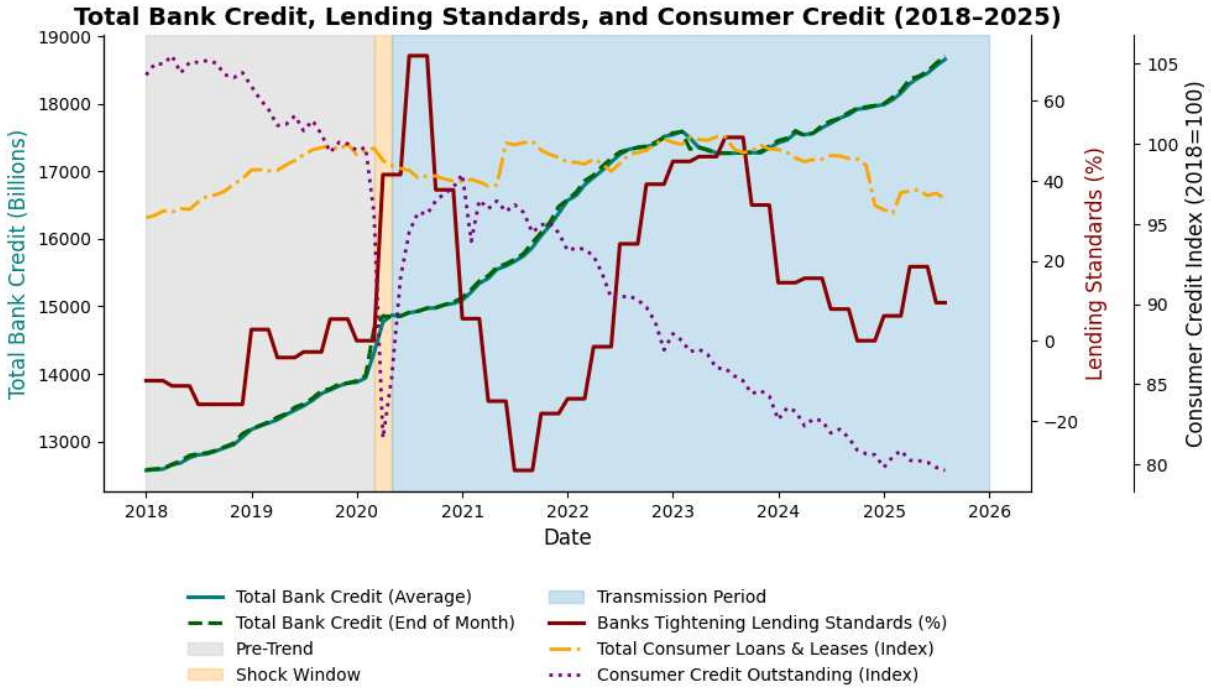


Figure 5. Total Bank Credit and Lending Standards

- Features related to consumer credit and bank lending over the 3 periods.

Manufacturing & Auto Production

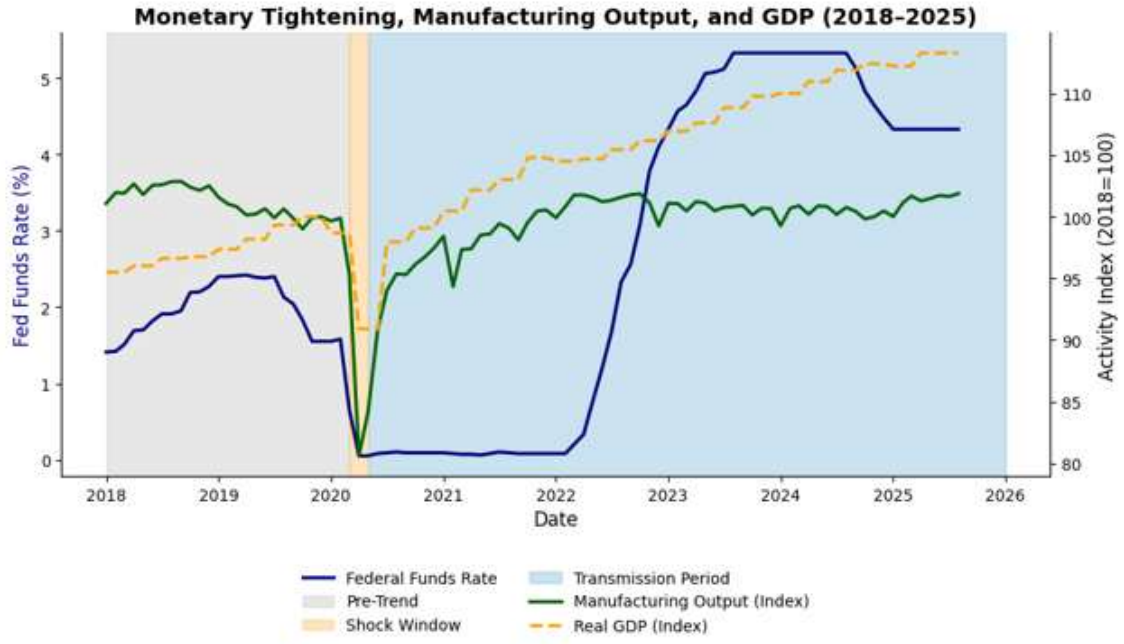


Figure 6. Monetary Tightening, Manufacturing Output, and GDP

- Monetary Tightening, Manufacturing Output, and GDP over the three periods.

Verification and Validation

- **Verification**
- Checked data accuracy and CPI use
- Confirmed frequency conversions
- Re-ran scripts for consistency
- Reviewed plots vs expected trends

- **Validation**
- Matched results with real events
- Confirmed objective alignment (2021–2025)
- Showed tightening effects across different sectors

Conclusions

Fed tightening after 2021 **raised borrowing cost**

Higher rates slowed auto sales and lending

Stricter credit cut consumer and business borrowing.

Manufacturing recovered but stayed below pre-COVID levels.

Credit shifted from easy to tight access

Fed policy **slowed inflation, stabilized demand**

Limitations

- Correlation ≠ Causation
- Not all features in FRED were used for analysis
- National-level data lacks regional detail
- Policy lag effect

Recommendations

- Adopt flexible rate policy
- Targeted credit support
- Encourage adaptable lending
- Improve Fed communication

